

## Power BI Sales Analytics Documentation

Comprehensive business intelligence documentation based on the submitted Power BI report.

### 1. Business Problem

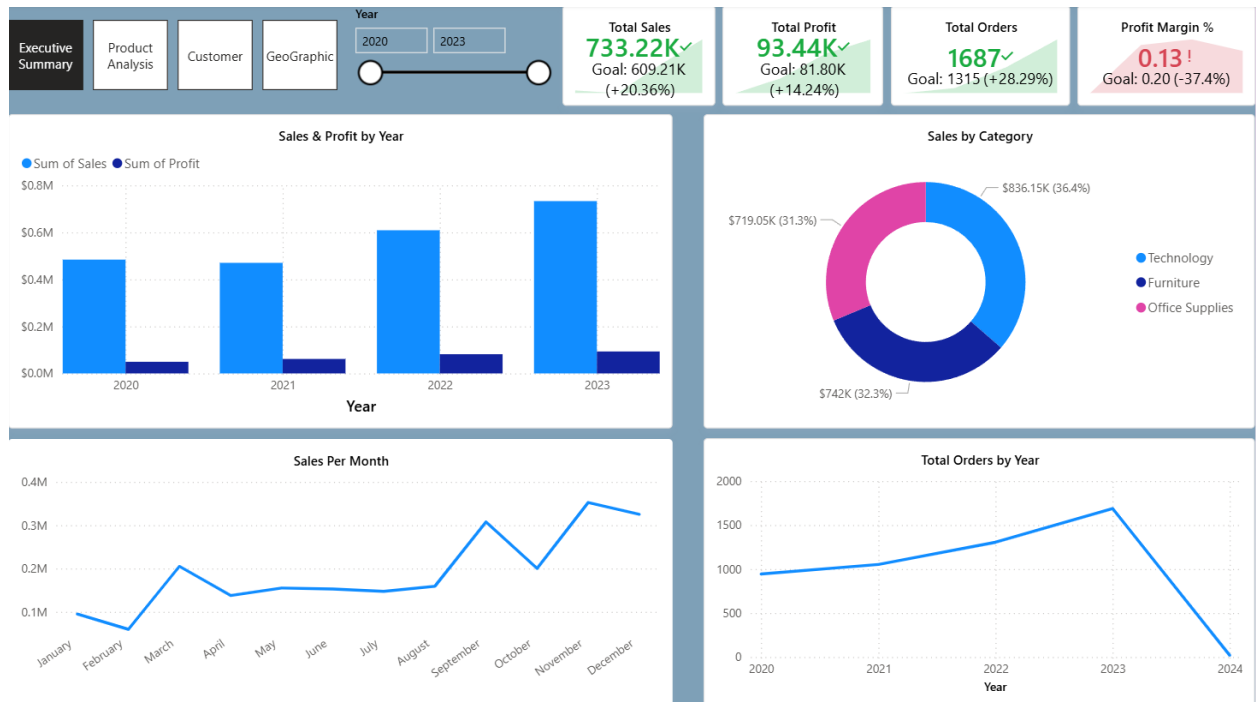
The company faces challenges in monitoring sales performance, profitability, customer purchasing behavior, product contribution, and geographic performance. Management requires a centralized analytical solution to identify high-performing products, profitable customers, regional opportunities, and year-over-year growth trends.

### 2. Dashboard Objectives

- Track sales, profit, orders, and margin performance.
- Compare performance across years.
- Identify best-performing product categories and customers.
- Analyze regional and geographic sales distribution.

### 3. Dashboard Screenshots and Explanations

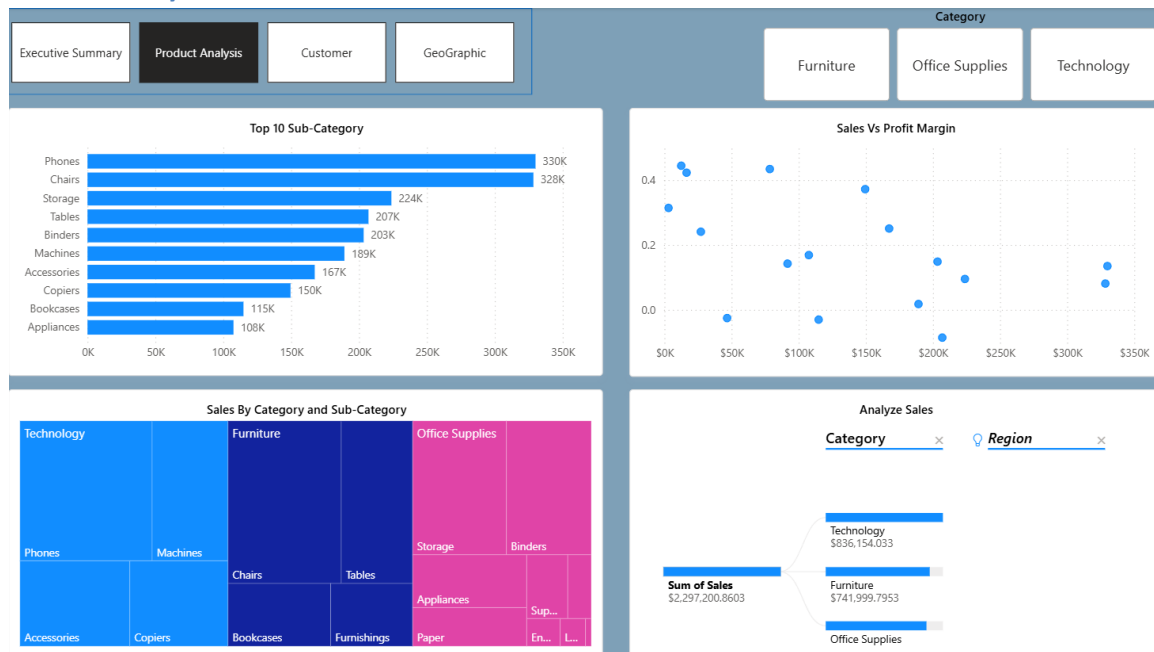
#### Executive Summary



This dashboard provides high-level KPIs such as Total Sales, Total Profit, Total Orders, and Profit Margin %. It includes yearly trend analysis, monthly sales trends, category sales

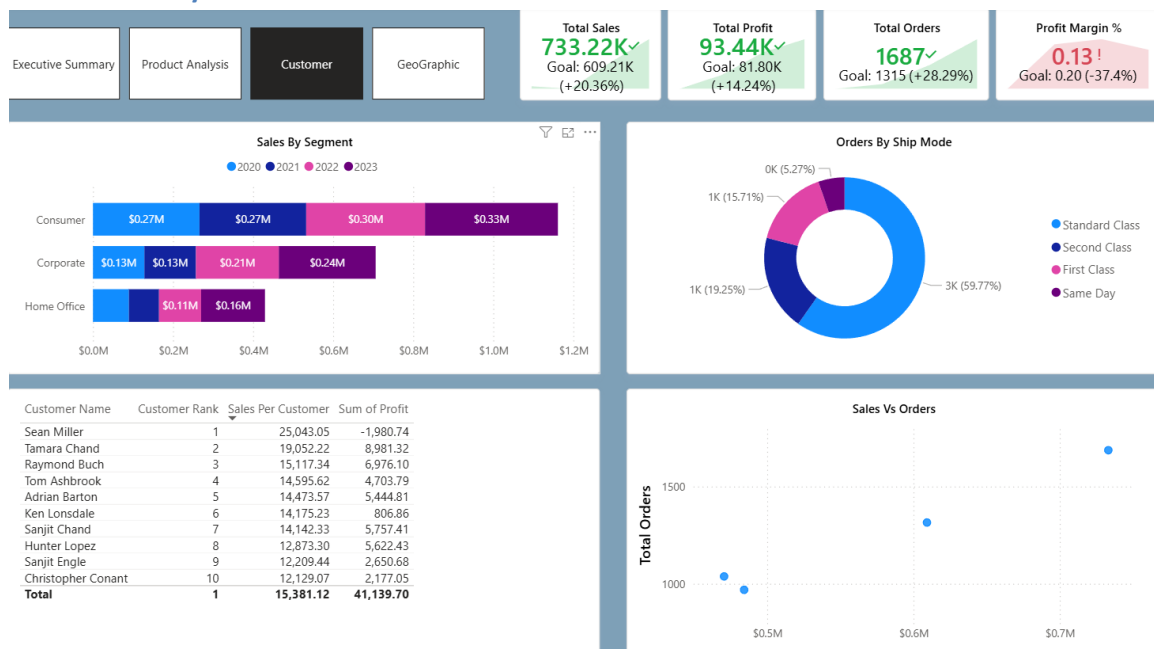
distribution, and order analysis. Decision makers can quickly monitor performance and compare outcomes against targets.

## Product Analysis



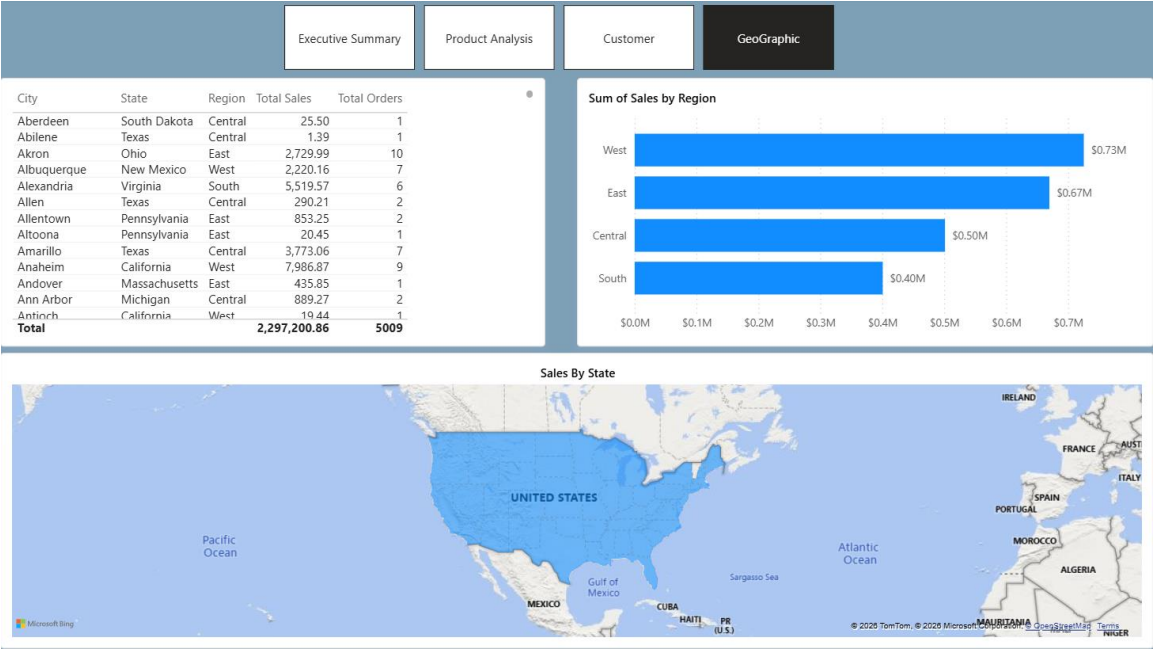
This page analyzes product performance by category and sub-category. The Top 10 Sub-Categories chart highlights best-selling items such as Phones and Chairs. Scatter plots compare sales vs profit margin while treemaps visualize category contribution. The decomposition tree helps drill down sales by category and region.

## Customer Analysis



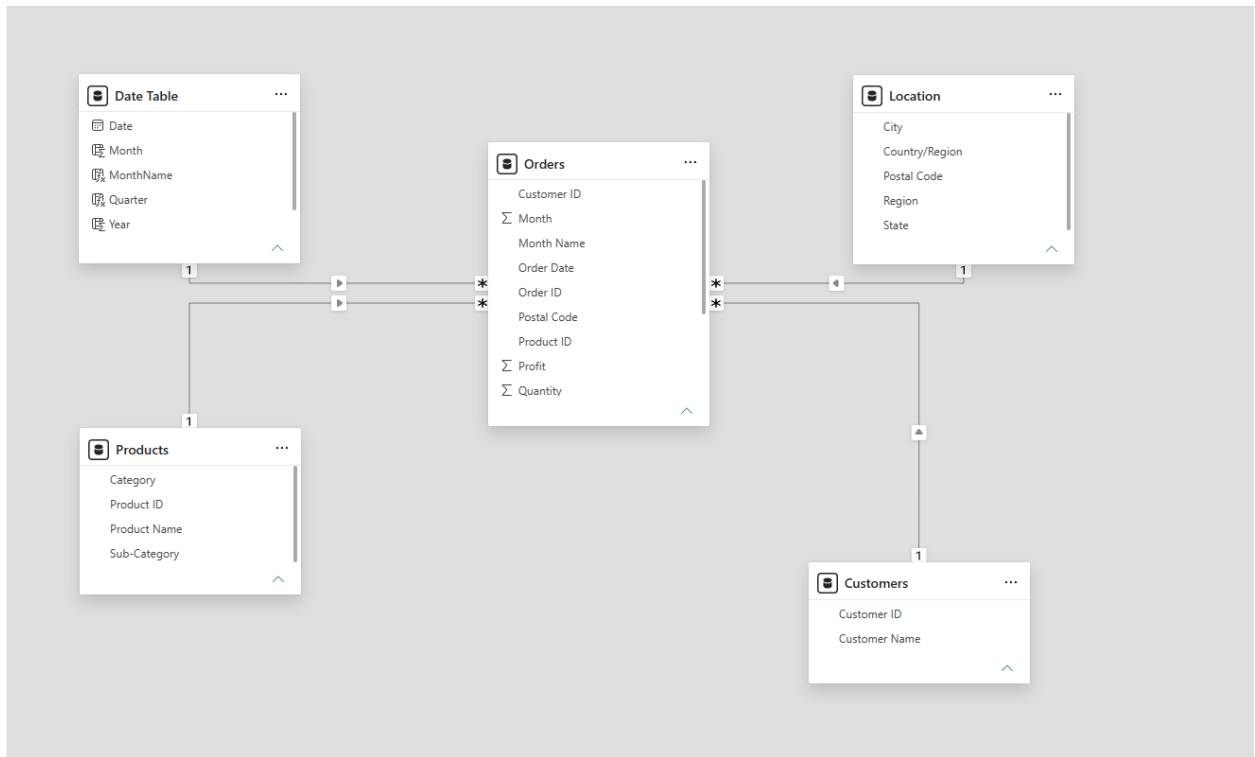
This page evaluates customer behavior and profitability. Segment analysis compares Consumer, Corporate, and Home Office customers. Ship mode analysis helps identify logistics preferences. Ranking tables identify top customers by sales contribution and profitability.

Geographic Analysis



This dashboard evaluates regional performance across the United States. West and East regions contribute the highest sales. The geographic map helps identify state-level performance and expansion opportunities.

## Data Model Relationships



The model follows a Star Schema design. Orders acts as the Fact Table connected to Date Table, Products, Customers, and Location dimensions using one-to-many relationships for efficient filtering and reporting.

## 4. Data Relationships

The data model follows a Star Schema. Orders is the central fact table containing transactional data such as Sales, Profit, Quantity, Product ID, Customer ID, and Postal Code. Dimension tables (Products, Customers, Location, Date Table) provide descriptive attributes.

## 5. DAX Measures Documentation

### Avg Order Value

```
Avg Order Value = DIVIDE([Total Sales], [Total Orders], 0)
```

DAX Formula: `DIVIDE([Total Sales],[Total Orders],0)`

Explanation: Calculates average revenue generated per order.

Business Purpose: This measure supports performance monitoring and business decision-making.

### Customer Rank

```
Customer Rank = RANKX(ALL(Customers[Customer Name]),[Total Sales],,Desc,DENSE)
```

DAX Formula: RANKX(ALL(Customers[Customer Name]),[Total Sales],,DESC,DENSE)

Explanation: Ranks customers by total sales.

Business Purpose: This measure supports performance monitoring and business decision-making.

#### Orders PY

---

Orders PY = CALCULATE([Total Orders], SAMEPERIODLASTYEAR('Date Table'[Date]))

DAX Formula: CALCULATE([Total Orders], SAMEPERIODLASTYEAR('Date Table'[Date]))

Explanation: Returns previous-year order count.

Business Purpose: This measure supports performance monitoring and business decision-making.

#### Profit Margin %

Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)

DAX Formula: DIVIDE([Total Profit],[Total Sales],0)

Explanation: Measures profitability ratio.

Business Purpose: This measure supports performance monitoring and business decision-making.

#### Profit Per Order

Profit Per Order = DIVIDE([Total Profit], [Total Orders], 0)

DAX Formula: DIVIDE([Total Profit],[Total Orders],0)

Explanation: Average profit generated per order.

Business Purpose: This measure supports performance monitoring and business decision-making.

#### Profit PY

---

Profit PY = CALCULATE([Total Profit], SAMEPERIODLASTYEAR('Date Table'[Date]))

DAX Formula: CALCULATE([Total Profit], SAMEPERIODLASTYEAR('Date Table'[Date]))

Explanation: Previous-year profit.

Business Purpose: This measure supports performance monitoring and business decision-making.

### Sales Per Customer

`Sales Per Customer = DIVIDE([Total Sales], [Total Customers], 0)`

DAX Formula: `DIVIDE([Total Sales],[Total Customers],0)`

Explanation: Average revenue per customer.

Business Purpose: This measure supports performance monitoring and business decision-making.

### Sales PY

`Sales PY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date Table'[Date]))`

DAX Formula: `CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date Table'[Date]))`

Explanation: Previous-year sales.

Business Purpose: This measure supports performance monitoring and business decision-making.

### Sales YoY Growth

`Sales YoY Growth = DIVIDE([Total Sales] - [Sales PY], [Sales PY], 0)`

DAX Formula: `DIVIDE([Total Sales]-[Sales PY],[Sales PY],0)`

Explanation: Year-over-year growth rate.

Business Purpose: This measure supports performance monitoring and business decision-making.

### Sales YoY Growth %

`Sales YoY Growth % = FORMAT([Sales YoY Growth], "0.0")`

DAX Formula: `FORMAT([Sales YoY Growth],"0.0")`

Explanation: Formatted YoY growth percentage.

Business Purpose: This measure supports performance monitoring and business decision-making.

### Top N Products Sales

`Top N Products Sales = CALCULATE([Total Sales], TOPN(10, ALL(Products[Product Name]), [Total Sales]))`

DAX Formula: `CALCULATE([Total Sales], TOPN(10, ALL(Products[Product Name]), [Total Sales]))`

Explanation: Calculates sales for top 10 products.

Business Purpose: This measure supports performance monitoring and business decision-making.

#### Total Customers

---

**Total Customers** = `DISTINCTCOUNT(Customers[Customer ID])`

DAX Formula: `DISTINCTCOUNT(Customers[Customer ID])`

Explanation: Counts unique customers.

Business Purpose: This measure supports performance monitoring and business decision-making.

#### Total Orders

**Total Orders** = `DISTINCTCOUNT(Orders[Order ID])`

DAX Formula: `DISTINCTCOUNT(Orders[Order ID])`

Explanation: Counts total orders.

Business Purpose: This measure supports performance monitoring and business decision-making.

#### Total Profit

**Total Profit** = `SUM(Orders[Profit])`

DAX Formula: `SUM(Orders[Profit])`

Explanation: Calculates total profit.

Business Purpose: This measure supports performance monitoring and business decision-making.

#### Total Quantity

**Total Quantity** = `SUM(Orders[Quantity])`

DAX Formula: `SUM(Orders[Quantity])`

Explanation: Calculates quantity sold.

Business Purpose: This measure supports performance monitoring and business decision-making.

### Total Sales

**Total Sales = SUM(Orders[Sales])**

DAX Formula: SUM(Orders[Sales])

Explanation: Calculates total revenue.

Business Purpose: This measure supports performance monitoring and business decision-making.

### 6. Business Insights

- Technology category generates the highest sales contribution.
- Phones and Chairs are the best-performing sub-categories.
- Consumer segment contributes the highest sales.
- West region outperforms other regions in revenue generation.
- Profit margin should be improved due to declining performance against goal.

### 7. Advises

- Focus marketing campaigns on high-performing categories.
- Improve profitability of low-margin products.
- Expand high-performing regions and investigate low-performing states.
- Develop customer retention strategies for top-ranked customers.